



Figure 3: Visualization of network flow results by varying the regularization coefficient ϵ for OFT. We observe that as ϵ decreases, our approximate solution calculated by EOFT-Sinkhorn approaches closer to the exact solution computed by Gurobi.

the marginal constraint. To address the issues mentioned above, we have modified the optimization objectives and constraints:

$$\min_{\mathbf{P}, \mathbf{q}} \langle \mathbf{D}, \mathbf{P} \rangle - \epsilon H(\mathbf{P}) \quad \text{s.t.} \quad \mathbf{P} \mathbf{1}_N = \mathbf{q} + \mathbf{d}, \mathbf{P}^\top \mathbf{1}_N = \mathbf{q} - \mathbf{s} + \mathbf{d}, \text{Diag}(\mathbf{P}) = \mathbf{d} \quad (5)$$

where $\mathbf{d} = d \cdot \mathbf{1}_N$ represents the virtual flow from each node to itself, given the constant d . Here, $\mathbf{D}$ is the distance matrix and we set $\mathbf{D}_{ii} = 0$ for every node i and increase the constraint $\mathbf{P}_{ii} = d$, ensuring that each node, including the isolated nodes mentioned earlier, carries at least d units of flow, which reduces the impact of isolated points for the solutions. Additionally, $\mathbf{q}$ represents the flow out of the nodes (excluding the virtual flow). Then given the net outflow $\mathbf{s}$, we can infer the flow into each node as $\mathbf{q} - \mathbf{s}$. The purpose of these transformations above is to establish a form similar to the marginal constraints in line with vanilla OT, thereby enabling the derivation of a Sinkhorn-like algorithm to obtain an approximate solution for OFT. Based on the method of Lagrange multipliers, we can derive the following proposition (see the proof in Appendix C.1), which aids us in further understanding the prosperity of the solution for OFT.

Proposition 2. *The solution to Eq. 5 is unique and has the form:*

$$\mathbf{P} = \text{Diag}(\mathbf{u}) \mathbf{K} \text{Diag}(\mathbf{v}) \quad \text{and} \quad \mathbf{q} = \frac{\mathbf{s}}{2} + \left((\mathbf{K}\mathbf{v}) \cdot (\mathbf{K}^\top \mathbf{u}) + \frac{\mathbf{s}^2}{4} \right)^{\frac{1}{2}} - \mathbf{d} \quad (6)$$

Here $\mathbf{u}, \mathbf{v}$ are two (unknown) scaling variables satisfying $\mathbf{u} \odot \mathbf{v} = \mathbf{1}$ and $\mathbf{K} = e^{(-\mathbf{C} + \text{diag}(\mathbf{h})) / \epsilon}$ where $\mathbf{h}$ is defined as:

$$\mathbf{h} = \epsilon (\log \mathbf{d} - \log \mathbf{u} - \log \mathbf{v}) \quad (7)$$

Similar to the entropic OT, the solution $\mathbf{P}$ still satisfies $\mathbf{P} = \text{diag}(\mathbf{u}) \mathbf{K} \text{diag}(\mathbf{v})$. However, the output flow $\mathbf{q}$ needs to be calculated, and the Gibbs kernel $\mathbf{K}$ is no longer a fixed matrix. Therefore, the iteration process appears more complex compared to the vanilla Sinkhorn algorithm, which we will discuss in detail next.

OFT-Sinkhorn Algorithm Here we propose the OFT-Sinkhorn algorithm 3.2, in which we aim to get the optimal solution of Eq. 5 through matrix-vector iterations. An intuitive idea is to iteratively update $\mathbf{P}$, $\mathbf{q}$, and $\mathbf{K}$ to obtain the optimal solution. For the coupling $\mathbf{P}$, based on the solution form $\mathbf{P} = \text{Diag}(\mathbf{u}) \mathbf{K} \text{Diag}(\mathbf{v})$ and the marginal constraints in Eq. 5, we derive the following iterations for $\mathbf{u}^{(l)}$ and $\mathbf{v}^{(l)}$ with the $(l+1)$ -th iteration:

$$\mathbf{u}^{(l+1)} = \frac{\mathbf{q}^{(l)} + \mathbf{d}}{\mathbf{K}^{(l)} \mathbf{v}^{(l)}}, \quad \mathbf{v}^{(l+1)} = \frac{\mathbf{q}^{(l)} - \mathbf{s} + \mathbf{d}}{(\mathbf{K}^{(l)})^\top \mathbf{u}^{(l+1)}} \quad (8)$$

where we initialize $\mathbf{v}^{(0)} = \mathbf{1}$, $\mathbf{K}^{(0)} = e^{-\mathbf{D}/\epsilon}$, and $\mathbf{q}^{(0)} = \max\{\tau, \tau + \mathbf{s}\}$ where τ is a sufficiently small positive value. By initializing in this manner, we can ensure that the two marginals of the

Algorithm 1 OFT-Sinkhorn: Sinkhorn-based Algorithm for Entropic Optimal Flow Transport

Input: Distance Matrix $\mathbf{D}$, Marginal Difference $\mathbf{s}$, maximum iteration number L , Virtual-flow $\mathbf{d}$.

Output: The Flow Matrix $\mathbf{P}^*$

Initialize $\mathbf{K} = \exp(-\frac{\mathbf{D}}{\epsilon})$ element-wise.

for $l = 0$ to $L - 1$ **do**

Update $u^{(l+1)}$ and $v^{(l+1)}$ according to Eq. 8

Update $\mathbf{K}^{(l+1)}$ by Eq. 10

Update $\mathbf{q}^{(l+1)}$ according to Eq. 9

end for

Compute $\mathbf{P} = \text{Diag}(\mathbf{u}) \cdot \mathbf{K} \cdot \text{Diag}(\mathbf{v})$

return $\mathbf{P}^* = \mathbf{P} - \mathbf{P}^\top$